

# StrideSense

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GPS-Powered Fitness Tracking · Flutter Mobile App

Built with Flutter · Powered by Supabase · Android & iOS

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# Why StrideSense Was Built

A group of university student athletes ran regularly but had no way to track their progress or stay accountable to each other.

Existing fitness apps were designed for general fitness tracking rather than the specific needs of run club communities - lacking features for group challenges, peer accountability, and club-based leaderboards.

Without proper tracking tools and social features built specifically for running groups, it was difficult to maintain motivation, measure improvement, and build team cohesion.

The goal: create a mobile app tailored specifically for run club communities, combining GPS tracking, progress visualization, and social accountability features - built for students, usable on any Android or iOS device.

## App Needs Identified

### Real-time GPS tracking

Distance, pace and route recorded live during a run

### Progress dashboard

Totals, history and weekly performance charts per user

### Social / community layer

Leaderboards, clubs and time-limited challenges

### Works offline

Local cache so the app functions without a network connection

→ StrideSense was designed specifically to meet all four needs

# What the App Had to Do

## REQUIREMENTS

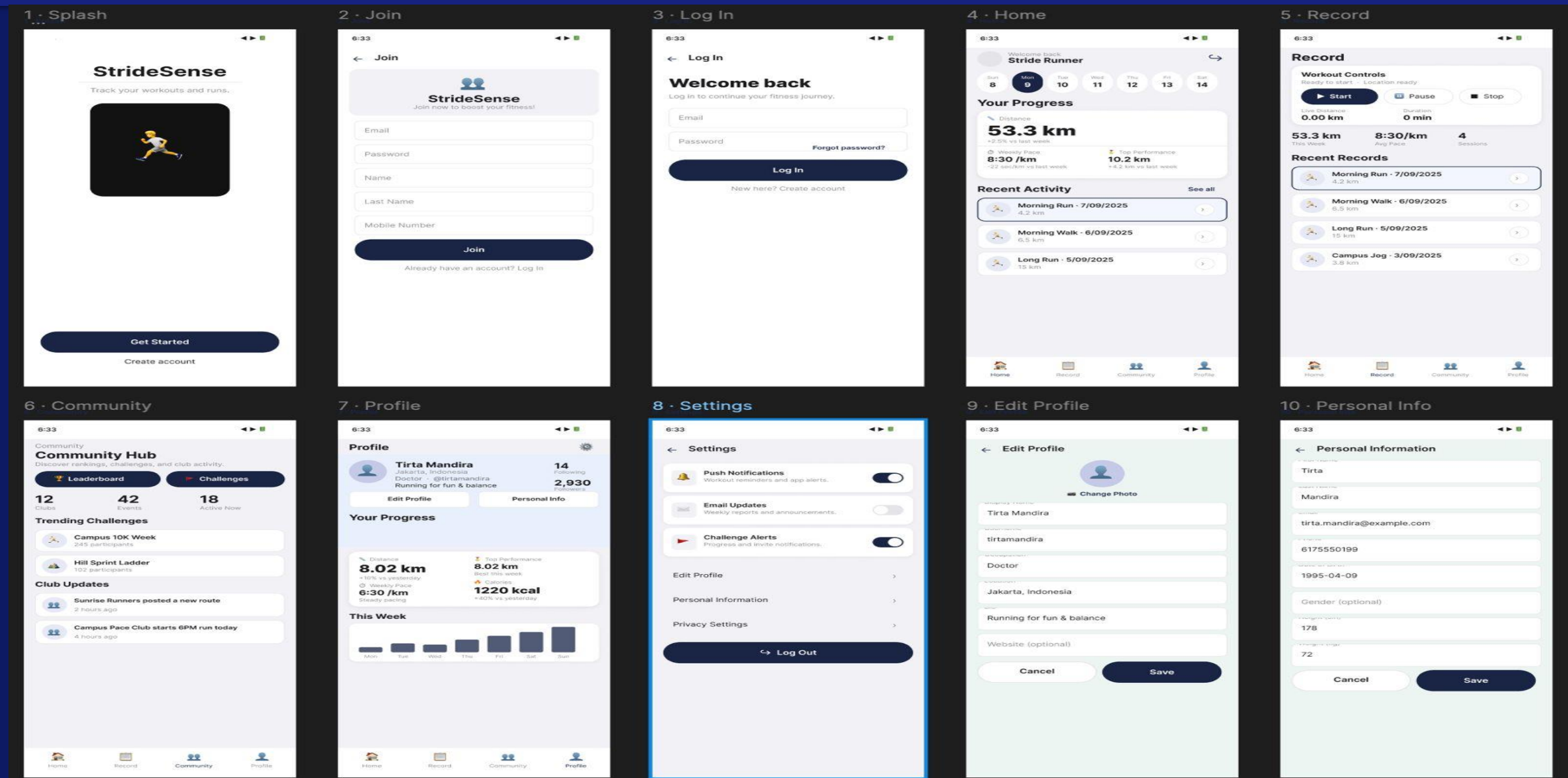
### Functional Requirements

- 1 User registration & login with email and password
- 2 Live GPS run tracking distance, pace and route
- 3 Automatic fallback to simulated GPS when unavailable
- 4 Workout history with map preview and summary
- 5 Profile editing including avatar photo upload
- 6 Browse and join clubs; view and join challenges
- 7 Leaderboard ranked by weekly or monthly distance

### Non-Functional Requirements

- 1 Offline-first local cache with background sync
- 2 Encrypted auth token storage on device
- 3 Sub-300 ms response for all local operations
- 4 Location access requires explicit user permission
- 5 Row Level Security enforced on all database tables
- 6 Supports Android API 21+ and iOS 14+

# Initial Wireframe Process — Figma



# Physical & Local Resources Used

DEVICE HARDWARE · PLATFORM APIs · ON-DEVICE STORAGE

## GPS / Location Sensor

Hardware: Device GPS chip (fine location)

Permission: ACCESS\_FINE\_LOCATION (Android)

Permission: CLLocationWhenInUse (iOS)

Package: geolocator ^11.1.0

Use: Live distance & route tracking during runs

Fallback: Simulated GPS path when unavailable

## Camera & Photo Library

Hardware: Device camera module

Permission: CAMERA (Android)

Permission: READ\_MEDIA\_IMAGES (Android)

Package: image\_picker ^1.1.2

Use: Capture or select profile avatar photo

Source: Camera shutter or device gallery

## Network / Internet

Hardware: WiFi adapter / Mobile data (4G/5G)

Permission: INTERNET (Android)

Package: supabase\_flutter, dio

Use: Supabase API calls (auth, DB, storage)

Use: OpenStreetMap tile loading for route maps

Offline: Cached local data used when unavailable

## Secure Device Storage

Hardware: Platform secure enclave

Android: Android Keystore System

iOS: Keychain / Secure Enclave

Package: flutter\_secure\_storage ^9.2.2

Stores: auth\_tokens\_v1 (encrypted)

Use: Secure session token persistence

## Local Key-Value Storage

Hardware: On-device flash storage

Platform: SharedPreferences (Android/iOS)

Package: shared\_preferences ^2.3.2

Stores: dashboard\_summary\_v1

Stores: profile\_data\_v1, sync\_queue\_v1

Use: Offline-first local data cache

## Map Tile Rendering

Hardware: GPU / device screen renderer

Package: flutter\_map ^6.1.0 + latlong2

Tiles: OpenStreetMap (via INTERNET)

Coords: Latitude/longitude from GPS chip

Use: Workout route map preview on log screen

Permission: INTERNET required for tile download

# Architecture & Module Assignments

5-LAYER ARCHITECTURE · TEAM MODULE OWNERSHIP

## L5 BACKEND / DATA

Supabase · PostgreSQL · Auth · Storage

## L4 DATA ACCESS

BackendApiClient · LocalStores · SyncWorker

## L3 SESSION & STATE

SessionController · SessionScope

## L2 NAVIGATION

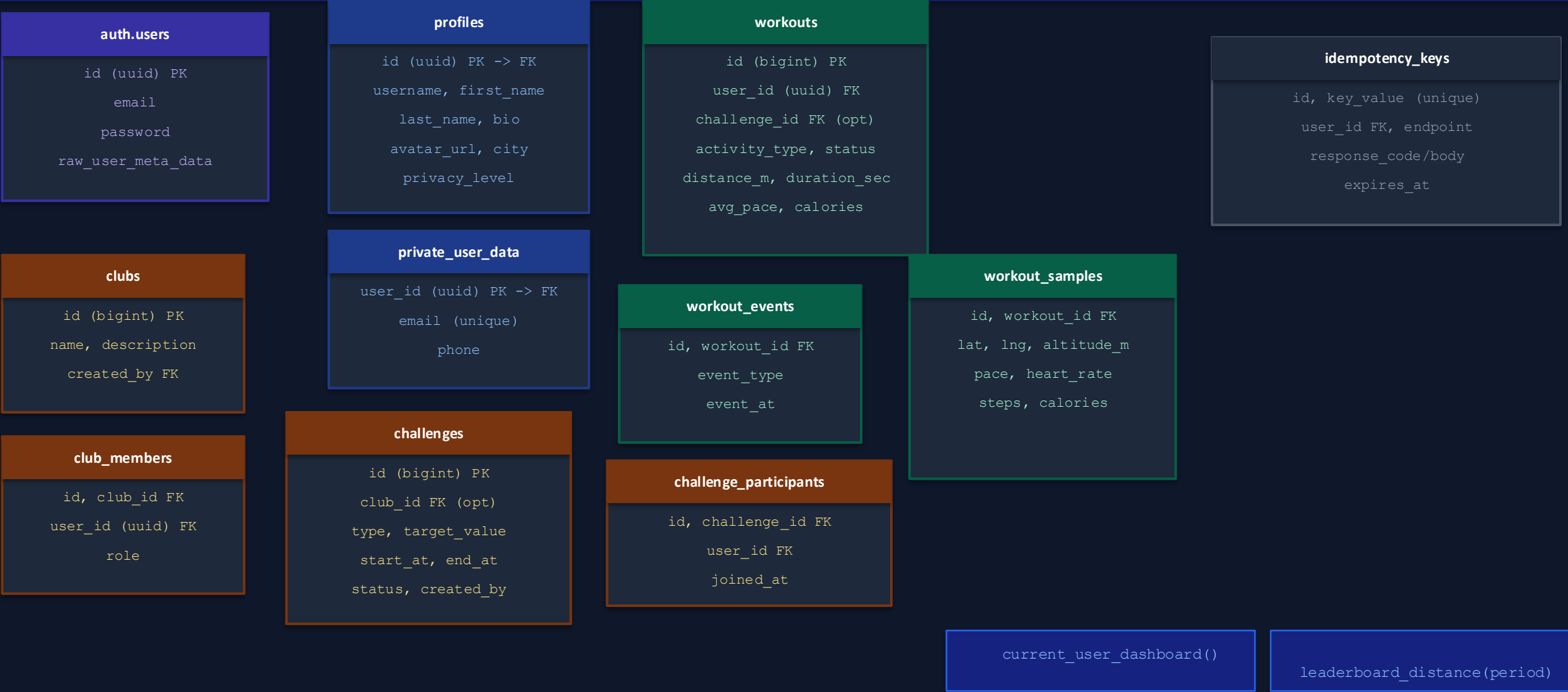
Routes · Auth Guard · Typed Arguments

## L1 PRESENTATION

Flutter Widgets · Material 3 · Navy Theme

| Module                 | File(s)                        | Member             |
|------------------------|--------------------------------|--------------------|
| Auth & Onboarding      | auth_screens.dart              | Yaa Asiamah        |
| Home Dashboard         | home_tab.dart                  | Yaa Asiamah        |
| Workout Recording      | home_tab.dart (Record section) | Yaa Asaimah        |
| Community Features     | community_tab.dart             | Shadrack Agyei Nti |
| Profile & Settings     | profile_tab.dart               | Ismail Adam        |
| Session / State / Data | session.dart                   | Shadrack Agyei Nti |
| Navigation & Routing   | lib/navigation/                | Ismail Adam        |
| PHP Backend / REST API | src/, public/, migrations/     | Ismail Adam        |
| Database Schema (SQL)  | supabase/migrations/           | Shadrack Agyei Nti |

# Entity Relationship Diagram



# Tools, Libraries, Frameworks & APIs

## FLUTTER APP TECH STACK

### Dart

Language

Primary language — all UI, logic and state management

### Flutter

SDK ^3.10.7

Cross-platform UI framework for Android & iOS from a single codebase

### Supabase

Backend-as-a-Service

Auth, Postgres DB, file Storage, RPC functions and Row Level Security

### PostgreSQL

via Supabase

Relational database with triggers, stored functions, enum types and RLS policies

### supabase\_flutter

^2.8.0

Dart SDK for Auth, database CRUD, RPC calls and avatar file upload

### geolocator

^11.1.0

Live GPS stream, distance computation and location permission checks

### flutter\_map + latlong2

^6.1.0 / ^0.9

Interactive workout route map rendering using OpenStreetMap tiles

### image\_picker

^1.1.2

Camera and photo gallery access for profile avatar photo upload

### flutter\_secure\_storage

^9.2.2

Encrypted auth token storage using device Keystore or Secure Enclave

### shared\_preferences

^2.3.2

Local key-value cache for dashboard, profile data and sync queue

### dio

^5.7.0

HTTP client used for structured error handling and offline fallback logic

### uuid

^4.5.1

Generates unique file path names for avatar uploads to Supabase Storage

# Deployment

BUILD · RELEASE · CONFIGURE

terminal

```
# Development run
flutter run \
  --dart-define=SUPABASE_URL=... \
  --dart-define=SUPABASE_ANON_KEY=...

# Release APK
flutter build apk --release \
  --dart-define=SUPABASE_URL=... \
  --dart-define=SUPABASE_ANON_KEY=...

# iOS release
flutter build ios --release
```

### SUPABASE SETUP

1. Create Supabase project
2. Apply SQL migration
3. Avatars bucket auto-created
4. Copy URL + anon key

### Android Android APK

flutter build apk --release · sideload or Play Store

### iOS iOS App

flutter build ios --release · TestFlight / App Store

### Cloud Supabase Cloud

Hosted Postgres + Auth + Storage · No self-hosting required

### Android Permissions Required

|                      |                        |          |
|----------------------|------------------------|----------|
| ACCESS_FINE_LOCATION | ACCESS_COARSE_LOCATION | INTERNET |
| READ_MEDIA_IMAGES    | CAMERA                 |          |

# How to Use StrideSense

## OPERATIONAL GUIDE

### Sign Up

- Tap Get Started on splash screen
- Fill email, password, name & phone
- Supabase creates profile automatically
- Lands on Home dashboard

### Log In

- Enter email and password
- Session stored securely on device
- Profile & dashboard loaded from cache
- Background sync fetches fresh data

### Record a Workout

- Open Record tab -> tap Start
- Grant location permission if prompted
- GPS tracks distance & time live
- Pause/Stop -> category -> Save

### View Progress

- Home: total distance & weekly pace
- Profile: weekly distance bar chart
- Tap any workout for route map + summary

### Community

- Leaderboard: weekly/monthly rankings
- Challenges: browse & join
- Clubs: browse, join or create

### Edit Profile

- Edit Profile — public display info
- Personal Info — private email & phone
- Change Photo — camera or gallery
- Saves locally, then auto-syncs